

Gaurav Singh

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Professional profile

Software developer and applied machine-learning researcher with current commercial experience building payment-platform software. I work across C#, .NET, TypeScript, Vue, REST APIs, Python, and modern machine-learning frameworks. My background spans computer science, engineering research, design, and a decade of university teaching. I have first-author peer-reviewed publications in applied machine learning and visual-mathematical literacy, and a Master of Information Technology focused on generative models for environmental time-series data.

Technical skills

Software engineering: C#, .NET, JavaScript, TypeScript, Vue, Azure DevOps, REST APIs, Git, and CI/CD; Docker and AWS.

Machine learning: supervised learning, feature engineering, model evaluation and cross-validation, time-series forecasting; scikit-learn, PyTorch, TensorFlow/Keras.

Deep learning: neural networks for computer vision, sequence modelling, and generative modelling.

Data and modelling: Python, NumPy, Pandas, SQL, Matplotlib; probability and statistical inference, exploratory data analysis, uncertainty quantification, nonlinear optimisation, techno-economic modelling, and longitudinal modelling.

LLMs and agents: LLM API integration, structured document extraction, agentic workflows, and evaluation.

Research interests

How representation choices shape what models can learn; making models work where data is sparse and incomplete; mathematical modelling of physical and economic systems; visual-mathematical literacy.

Education

2023-2025	Master of Information Technology Griffith University, Gold Coast, Australia; GPA 6.31/7 Thesis: "Enhancing Climate Forecasting with Advanced Generative Models for Synthetic AQI Data", supervised by Dr. Elizabeth Chang.
2013-2015	Advanced Diploma, Innovation & Experience Design Srishti Institute of Art, Design and Technology, Bangalore, India
2008-2012	B.Tech, Computer Science & Engineering JNTU Anantapur, India; First Class with Distinction

Experience

2025-present	Software Developer, Payments Technology Company, Australia Develop payment-platform software supporting recurring billing, direct debit, subscriptions, bank-account runs, card payments, hosted payment pages, and API integrations. Current stack: C#, .NET, TypeScript, Vue, Azure DevOps, and REST APIs.
2022-2023	Research Assistant, Griffith University, Australia Applied machine learning, computer vision, and image processing to engineering research on deformation mechanisms in re-entrant honeycomb auxetics. The work contributed to a first-author peer-reviewed publication.
2018-2022	Design Copilot, Topcoder, Remote Ran design challenges for Fortune 500 clients and guided distributed groups of designers through competition briefs, after several years competing on the platform.

2017-2022	Faculty Member and Researcher, Srishti Manipal Institute, Bangalore, India Headed the Master of Design in Design Computation, founded the Experimental Maths Lab, and taught programming, algorithms, machine learning, physical computing, interaction design, and human-computer interaction.
2008-present	Founder and Researcher, Mathscapes, India / Remote Develop open-source libraries and design tools focused on algorithms, recreational mathematics, and accessible mathematical practice, including the Iterflow streaming-statistics library.

Publications

A Framework for Visual-Mathematical Literacy in Applied Machine Learning: Why Representation Choices Shape What Models Can Learn. G. Singh, R. S. Dhari. IOP Machine Learning, 2026. [doi:10.1088/3049-4761/ae7df3](https://doi.org/10.1088/3049-4761/ae7df3)

Techno-economic pathways modeling and nonlinear optimized SEEA-ROI longitudinal dynamic simulation for decarbonising Australian heavy transportation systems. G. Singh, E. Chang, Y. Karaca. Fractals, 2026. [doi:10.1142/S0218348X26400633](https://doi.org/10.1142/S0218348X26400633)

Automated detection of deformation mechanisms in re-entrant honeycomb auxetics using machine learning. G. Singh, R. S. Dhari, Z. Javanbakht. International Journal of Protective Structures, 2024. [doi:10.1177/20414196241281069](https://doi.org/10.1177/20414196241281069)

ReRide: A Bike Area Network for Embodied Self-monitoring during Motorbike Commute. N. Bagalkot, G. Singh, V. Rath, T. Sokoler, A. Shukla. Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '19), 443-450, 2019. [doi:10.1145/3294109.3300986](https://doi.org/10.1145/3294109.3300986)

ReRide. N. Bagalkot, T. Sokoler, R. Shaikh, G. Singh, A. E. Lillie, P. Dixit, A. Rai, V. Chakravarthy, A. Senthil. Human-Computer Interaction - INTERACT 2017. Lecture Notes in Computer Science, vol 10516. Springer, Cham, 2017. [doi:10.1007/978-3-319-68059-0_43](https://doi.org/10.1007/978-3-319-68059-0_43)

SnapTag: Leveraging Situated Memory to enhance self-efficacy for well-being. S. Baadkar, G. Singh, A. Saraf, N. Bagalkot. Proceedings of the India HCI 2014 Conference on Human-Computer Interaction, 136-141, 2014. [doi:10.1145/2676702.2676719](https://doi.org/10.1145/2676702.2676719)

Selected software

Iterflow: Composable Streaming Statistics for JavaScript. G. Singh. Mathscapes, 2026. [doi:10.5281/zenodo.18610143](https://doi.org/10.5281/zenodo.18610143)

Teaching and academic leadership

Designed and taught 30 courses and workshops over a decade (2013-2022), mostly at Srishti Manipal Institute in Bangalore, spanning creative coding, algorithmic botany, physical computing, machine learning, interaction design, and human-computer interaction.

Awards

2024	Griffith Award for Academic Excellence · Griffith University
2024	STEM Professional Gold Achievement · Griffith University
2023	Adobe Fund for Design · Design grant
2021	Adobe Fund for Design · Design grant
2017	Topcoder Open Design Champion · India Regionals
2012	Topcoder Open Studio Finals · 6th, Orlando
2011	Topcoder Open Studio Finals · 11th, Fort Lauderdale

Service

2018	Demonstrations track co-chair, IndiaHCI 2018 · Bangalore, India
2017	Cumulus Conference, organising team · Bangalore, India

Supervision

2020	Discovering Algorithmic Literacy through Explainable AI · D. Dixit, S. Singh
2020	Financial indicators that motivate impact investment · A. Avadhana, N. Patny, S. Mishra
2013	Location and arrival prediction · Shobha R., Pavani A., Kalpana, JNTU Anantapur